Ans-Ex4

1. $\phi_{11} = 3/7, \phi_{22} = 2/9$. Note that $\phi_{11} = \rho_1, \phi_{22} = (\rho_2 - \rho_1^2)/(1-\rho_1^2)$.

2. For ARMA(1,1) process, i.e.,

$$y_t = \alpha y_{t-1} + \beta\varepsilon_{t-1} + \varepsilon_t,$$

its ACF is given by

$$\rho_1 = (1+\alpha\beta)(\alpha+\beta)/(1+\beta^2+2\alpha\beta) = -0.746,$$
$$\rho_k = \alpha\rho_{k-1}, k = 2,3,...$$

hence you can plot ρ_k.

3. Possible model is $\nabla X_t = a_t$.

The obvious problem here is the sample size is too small. One thing that should be done is to increase the sample size.

4.. MA(2) is reasonable.

5. (a) B=2 or 5, which are larger than 1. Hence time series is stationary.

(b) Because $(1-0.5B)(1-0.2B)X_t = Z_t$

Coef. of B^4 is 0.1031.

6. (i) $|\phi| < \dfrac{-1+\sqrt{5}}{2}$

(ii)

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + a_t$$
$$\Rightarrow X_t X_{t-j} = \phi_1 X_{t-1} X_{t-j} + \phi_2 X_{t-2} X_{t-j} + a_t X_{t-j},$$

Consider j=0, 1, 2 and take expectation. $\phi = 0.509, \sigma^2 = 2.98$

7. ARMA(1,1), bookwork..

8. MA(2)